Security principles and threat modelling



Cyber security · University · Lecture 1

In this lecture

- Apply classic design principles.
- Threat-model with STRIDE.
- Teach principles through a real system.

Remember

- Deny by default.
- Least privilege.
- STRIDE every flow.

Model the portal

1. Warm-up (5 min): What went wrong?
2. Teach (10 min): Principles.
3. Draw (15 min): The data flow.
4. Threats (25 min): STRIDE.
5. Wrap-up (5 min): One-minute paper.

Let’s discuss

1. Which threat surprised you?
2. Which principle covers most threats?
3. How will you grade the threat table?

Exit ticket

- Name one trust boundary in a system you use and one STRIDE threat at it.